

Comparison and Analysis of Stiefel Manifold Optimizers in Deep Learning Applications

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Abstract

With the increasing popularity of large-scale LLMs and deep neural nets, there has been an increasing interest in designing the optimization algorithm to incorporate the inherent matrix/tensor structure of the model into its updates. One of the most successful of these approaches has been Muon (Jordan et al.), which approximates the momentum on the Stiefel manifold. However, other approaches have proposed updates which keep the position constrained to the manifold, namely the StiefelAdam optimizer (Tao et al.) There has not been a thorough analysis of the tradeoff between Stiefel position/momentum or the decision boundary between these two approaches. In this work, we first perform a rigorous analysis of Muon and StiefelAdam on a quadratic minimization problem, identifying a decision boundary and categorization of saddle points for StiefelAdam. We then investigate the updates on a simplified attention mechanism, identifying that StiefelAdam’s failure mode from the quadratic problem generalizes. Finally, we propose a new optimizer which applies the Stiefel constraint to both position and momentum.

Optimizers

First for notation, we define the Stiefel manifold as $St(n, m) := \{\mathbf{X} \in \mathbb{R}^{n \times m} : \mathbf{X}^T \mathbf{X} = \mathbf{I}_{m \times m}\}$ for $n \geq m$. We say a weight matrix $\mathbf{W} \in \mathbb{R}^{n \times m}$ is constrained to the Stiefel manifold if, throughout training, the matrix $\mathbf{W}_t \in St(n, m)$ as enforced by the StiefelAdam optimizer.

Algorithm 1 Muon Optimizer

learning rate η

Update rule for gradient $\mathbf{G}$ at time $t+1$:

$$\mathbf{W}_{t+1} \leftarrow \mathbf{W}_t - \eta \text{msign}(\text{EMA}(\mathbf{G}))$$

Algorithm 2 StiefelAdam

$\mathbf{W}_0 \in St(n, m)$, learning rate η , β_1, β_2 , ϵ
Compute Euclidean gradient:

$$\mathbf{G}_t \leftarrow \nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}_t)$$

Decompose gradient into tangent components:

$$\mathbf{f}_t \leftarrow \frac{1-b}{2}(\mathbf{W}_t^T \mathbf{G}_t - \mathbf{G}_t^T \mathbf{W}_t), \mathbf{g}_t \leftarrow (\mathbf{I} - \mathbf{W}_t \mathbf{W}_t^T) \mathbf{G}_t$$

Update first moments:

$$\mathbf{Z}_{t+1}(\mathbf{f}_t), \mathbf{U}_{t+1}(\mathbf{g}_t)$$

Update second moments:

$$\mathbf{p}_{t+1}(\mathbf{f}_t^{\odot 2}), \mathbf{q}_{t+1}(\mathbf{g}_t^{\odot 2})$$

Take tangent step:

$$\mathbf{W}^\dagger \leftarrow \mathbf{W}_t + \eta \left(\mathbf{W}_t \frac{\mathbf{Z}_{t+1}}{\mathbf{p}_{t+1}} + \frac{\mathbf{U}_{t+1}}{\mathbf{q}_{t+1}} \right)$$

Retract onto the Stiefel manifold:

$$\mathbf{W}_{t+1} \leftarrow \mathbf{W}^\dagger [(\mathbf{W}^\dagger)^T \mathbf{W}^\dagger]^{-1/2}$$

Quadratic Model

$$\min_{\mathbf{W} \in \mathbb{R}^{n \times n}} \|\mathbf{A}\mathbf{W} - \mathbf{A}\|_F^2 \quad (1)$$

is the quadratic minimization problem, where we evaluate models with various settings of $\mathbf{A}$ based on the condition number $\kappa(\mathbf{A}) = \frac{\sigma_1}{\sigma_n}$, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$ the ordered singular values of $\mathbf{A}$.

Quadratic Experiments

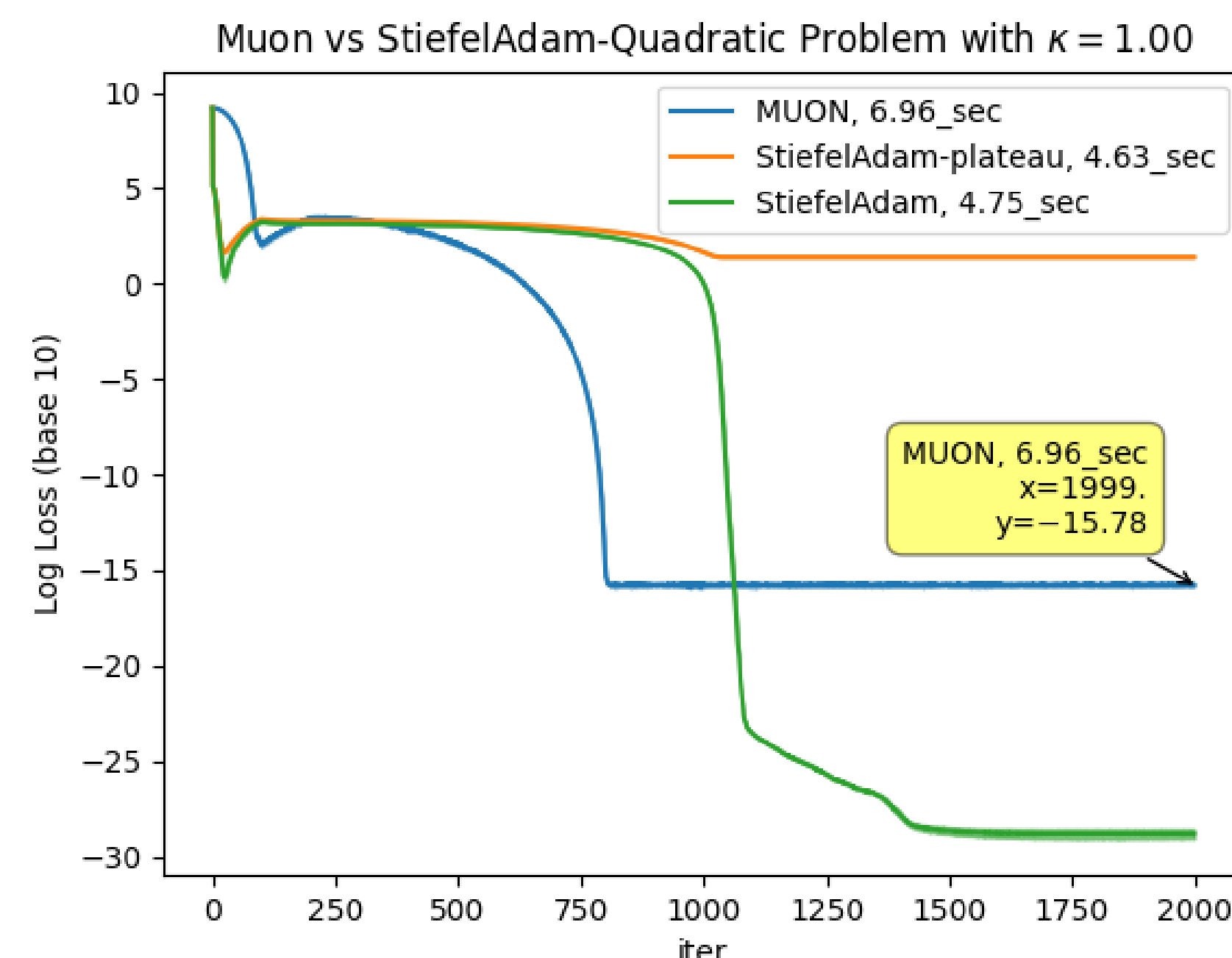


Figure 1: Muon vs StiefelAdam on well-conditioned Quadratic problem (1), 10 runs on different prefactors $\mathbf{A}$

Figure 1 shows how StiefelAdam experiences barren plateaus but can outperform Muon for well-conditioned prefactors.

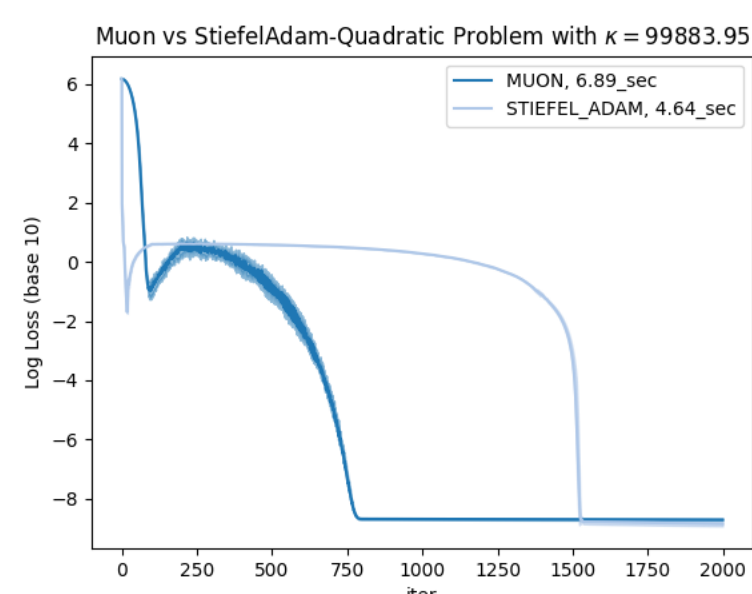


Figure 2

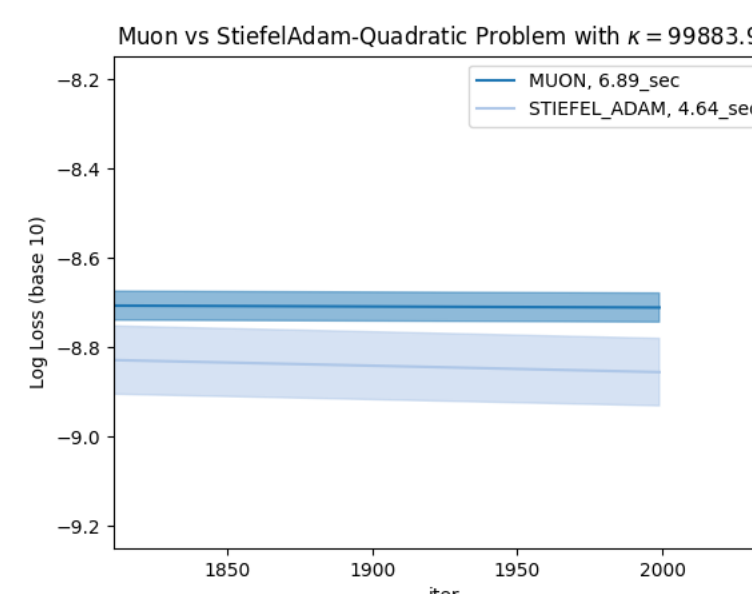


Figure 3

Figure 2 demonstrates how larger condition number avoids barren plateaus for StiefelAdam.

Quadratic Analysis

We can understand the barren plateau by analyzing StiefelAdams’s update under the well-conditioned case. With $\kappa(\mathbf{A}) = 1$ this implies $\mathbf{A}$ is orthogonal. So evaluating the gradient of (1), we see $\mathbf{G} = 2\mathbf{A}^T \mathbf{A}(\mathbf{W} - \mathbf{I}) = 2(\mathbf{W} - \mathbf{I})$. Now StiefelAdam’s update depends on $\mathbf{X}^T \mathbf{G} - \mathbf{G}^T \mathbf{X}$, which evaluated for this model gives $2(\mathbf{W} - \mathbf{W}^T)$. So StiefelAdam is dependent upon the skew component of $\mathbf{W}$, and experiments showed the plateau occurred when the weight became symmetric.

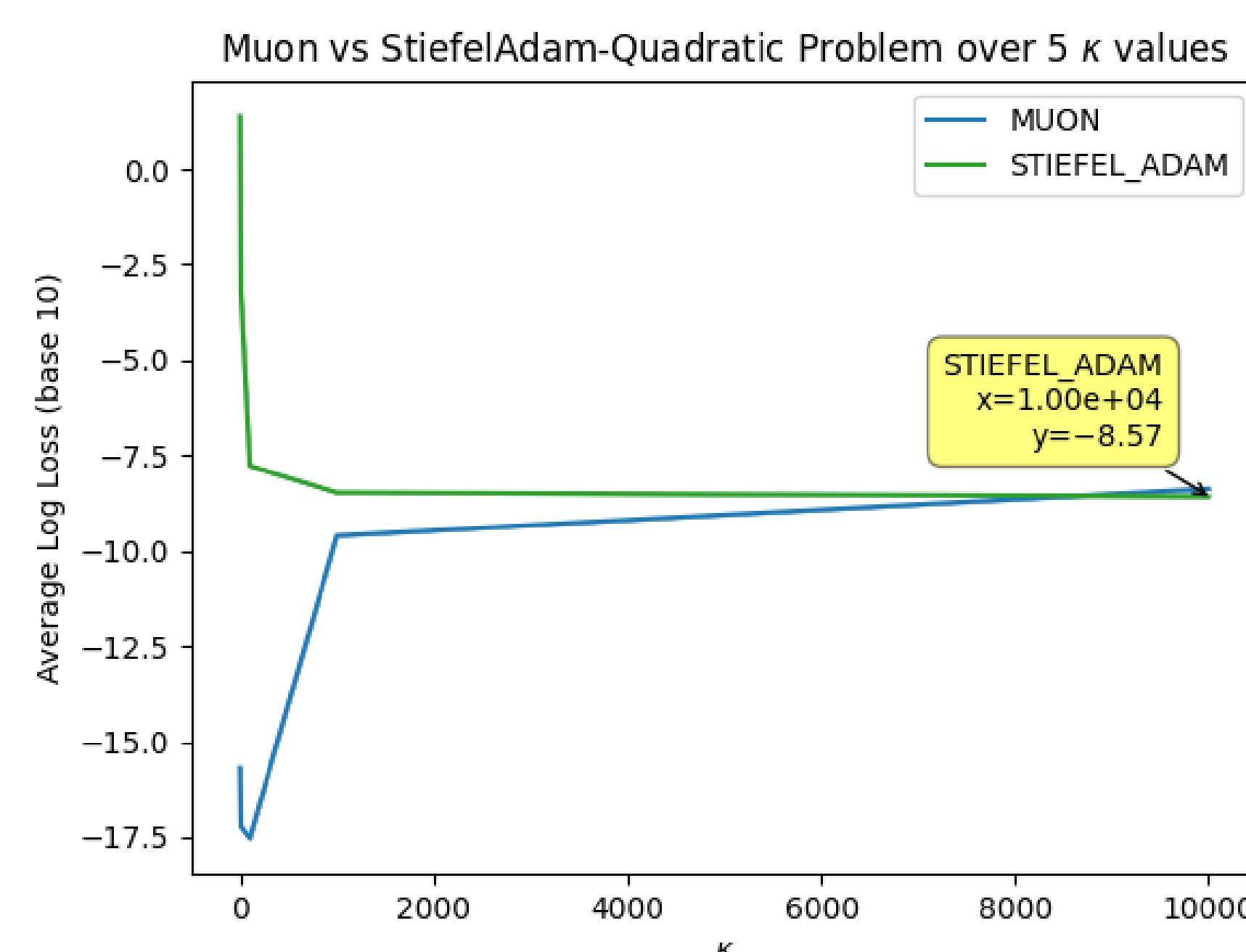


Figure 4: StiefelAdam outperforms Muon with lower average loss as the condition number increases.

Transformer Analysis

Having established the baseline behavior of Muon and StiefelAdam on (1), we now investigate their training dynamics on a simplified transformer model. We define this simplified model as

$$\begin{aligned} \mathbf{X} &\in \mathbb{R}^{T \times d_{\text{model}}}, \mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d_{\text{model}} \times d_k}, \\ \mathbf{W}_V &\in \mathbb{R}^{d_{\text{model}} \times d_v}, \mathbf{W}_O \in \mathbb{R}^{d_v \times d_{\text{model}}} \\ \mathbf{A} &= \text{Attention}(\mathbf{X}\mathbf{W}_Q, \mathbf{X}\mathbf{W}_K, \mathbf{X}\mathbf{W}_V)\mathbf{W}_O \\ \mathbf{S} &= \text{Softmax}(\mathbf{A}) \\ \mathcal{L}(\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V, \mathbf{W}_O) &= \mathcal{L}_{CE}(\mathbf{S}, \mathbf{T}) \end{aligned} \quad (2)$$

In Tao et al., they propose applying StiefelAdam to the subset of weights $\mathbf{W}_Q, \mathbf{W}_K$ and keeping the remaining matrices euclidean with Muon optimization. For fair comparison, we must determine if these two settings permit the same minimization solution.

It can be proved that constraining only one of $\mathbf{W}_Q, \mathbf{W}_K$ does not change the minimizer landscape for the transformer model. Specifically, for $\mathbf{W}_Q \in St(d_{\text{model}}, d_k)$ with optimal unconstrained weights $\tilde{\mathbf{W}}_K, \tilde{\mathbf{W}}_Q$, set

$$\mathbf{W}_Q = \mathbf{Q}_{\text{decomp}}, \mathbf{W}_K = \tilde{\mathbf{W}}_K \tilde{\mathbf{W}}_Q^T \mathbf{Q}_{\text{decomp}}^T \quad (3)$$

for the tall QR decomposition of $\tilde{\mathbf{W}}_Q = \mathbf{Q}_{\text{decomp}} \mathbf{R}$.

Now for the dual $\mathbf{W}_Q, \mathbf{W}_K$ Stiefel constrained case. As might be expected, the Stiefel constraint on the attention logits means we can find unattainable minimizers. However, these are largely pathological cases which wouldn’t be encountered in a live training setup.

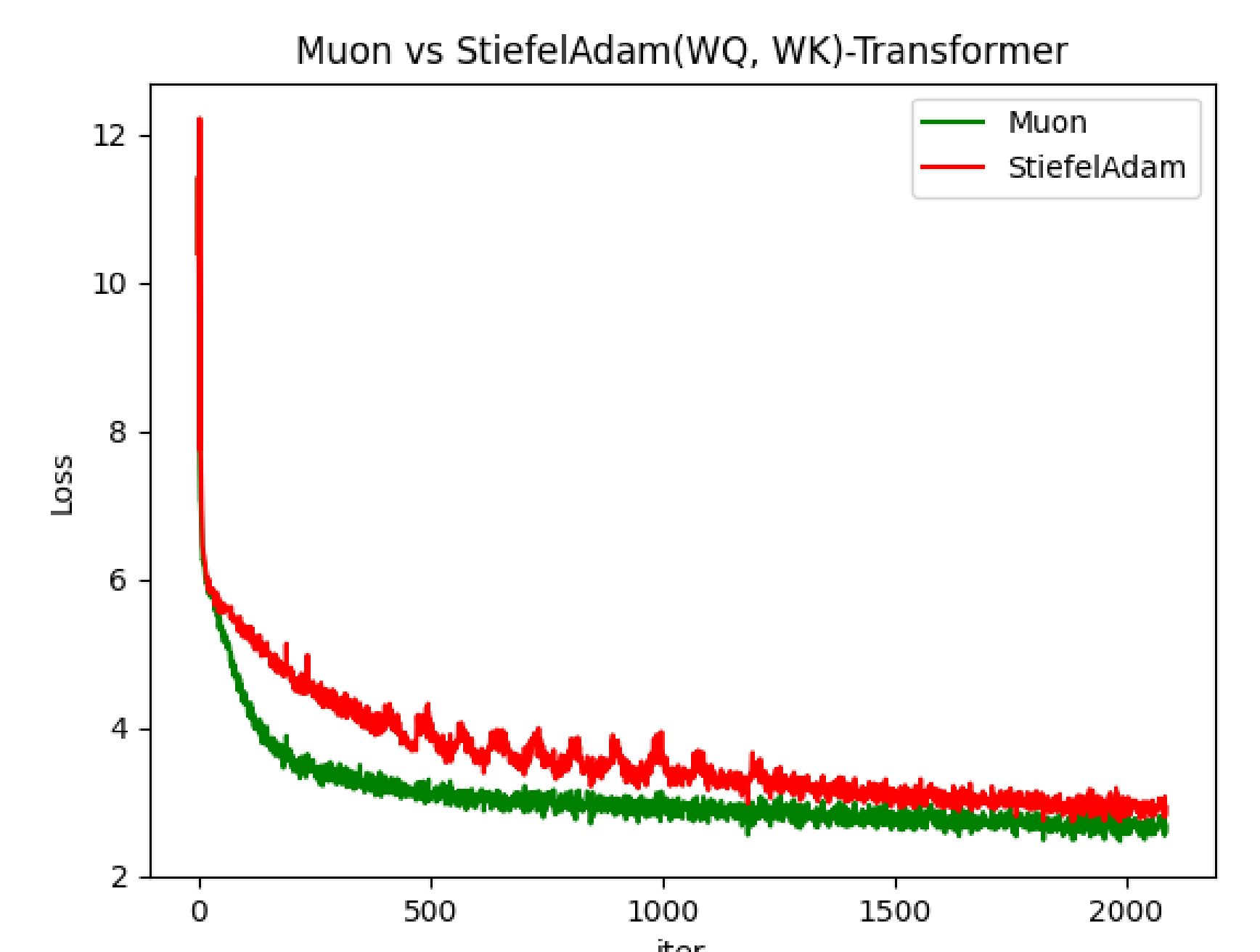


Figure 5: Muon vs StiefelAdam (only $\mathbf{W}_Q, \mathbf{W}_K \in St(d_{\text{model}}, d_k)$) on nanochat, notice higher variance of StiefelAdam.

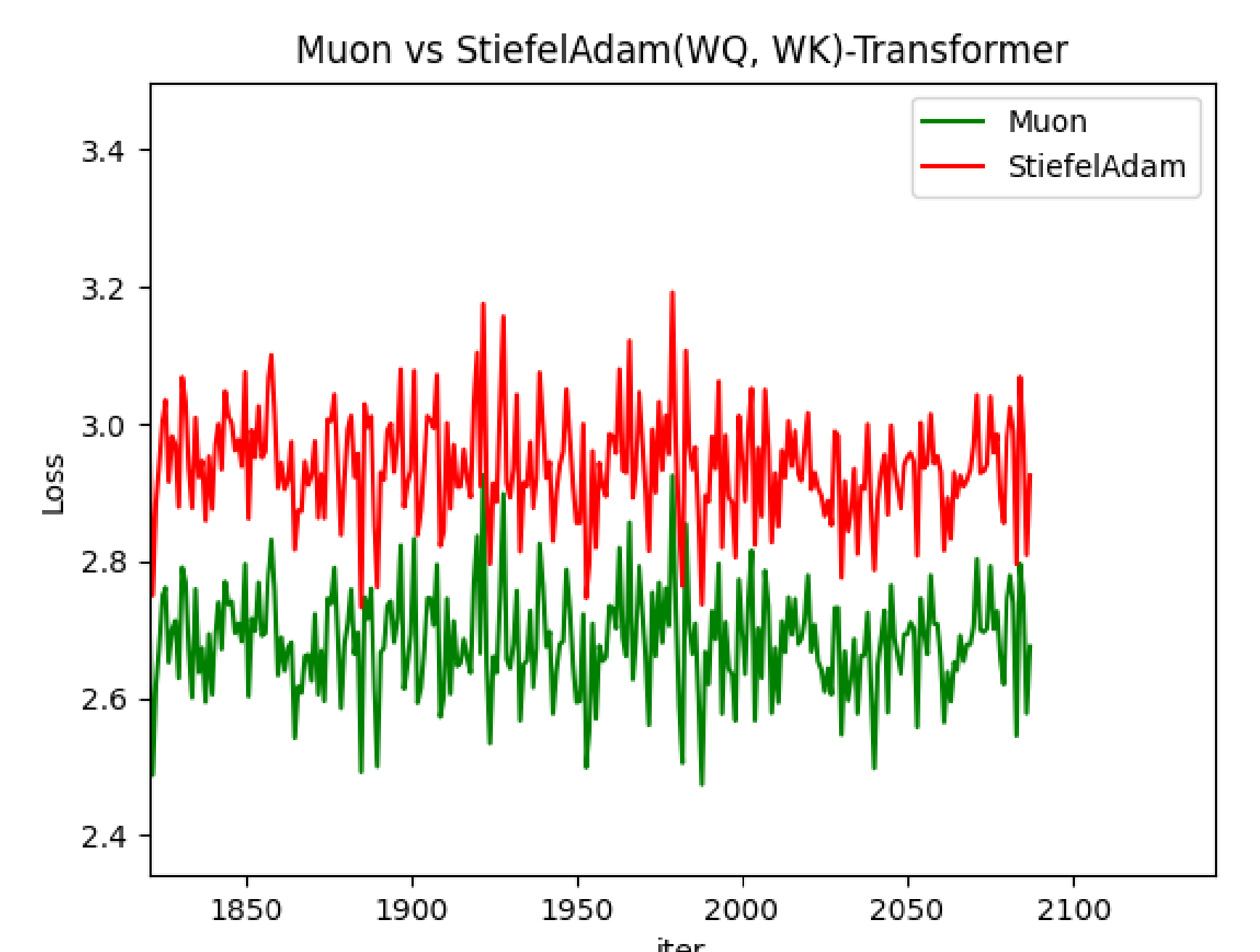


Figure 6: Despite higher variance of StiefelAdam, ultimately the final losses are similar.